

The Cosmovirus as a Category

Objects, Morphisms, and the Ouroboros Endofunctor

Cosmovirus Formalization Project

September 2, 2026

Abstract

We recast the $\mathbf{E}_8^* \times \varphi$ -SiS₂/HPV16 Cosmovirus framework as a small category *Cosmo* whose objects are the descent layers and whose morphisms are the framework's operators. We state the composition and identity laws, exhibit the descent as a directed path, and show that the Ouroboros loop is an endofunctor whose associated “feedback” square commutes up to a natural transformation.

1 The category *Cosmo*

Definition 1.1 (Objects). The object class $\text{Ob}(\text{Cosmo})$ consists of the six descent layers

$$\text{Ob}(\text{Cosmo}) = \{ \mathbf{E}_8^*, P_\varphi, \text{SiS}_2, \text{Tri}, \text{HPV16}, \Omega \},$$

where $\mathbf{E}_8^*$ is the pure-symmetry source, P_φ the golden-scaled layer, SiS₂ the material substrate, Tri the triality branch point, HPV16 the infected oncocode layer, and Ω the Ouroboros loop state.

Definition 1.2 (Generating morphisms). The category is generated (under composition) by the arrows

$$\begin{aligned} s : \mathbf{E}_8^* &\rightarrow P_\varphi && (\varphi\text{-SCL golden projection}) \\ m : P_\varphi &\rightarrow \text{SiS}_2 && (\text{materialization / substrate cast}) \\ t : \text{SiS}_2 &\rightarrow \text{Tri} && (\text{triality } \text{SO}(8) \rightarrow \text{Spin}(8)) \\ h : \text{Tri} &\rightarrow \text{HPV16} && (\text{HPV16 infection}) \\ r : \text{HPV16} &\rightarrow \Omega && (\varphi^{101} \text{ recursion}) \\ \ell : \Omega &\rightarrow \mathbf{E}_8^* && (\text{Ouroboros loop-back}) \end{aligned}$$

together with the identity id_X for each object X .

Axiom 1.3 (Composition and identities). *For composable arrows $f : X \rightarrow Y$, $g : Y \rightarrow Z$ the composite $g \circ f : X \rightarrow Z$ exists and is associative; each id_X is a two-sided unit: $f \circ \text{id}_X = f = \text{id}_Y \circ f$. In particular the descent morphism*

$$\Psi = \ell \circ r \circ h \circ t \circ m \circ s : \mathbf{E}_8^* \rightarrow \mathbf{E}_8^*$$

is a well-defined endomorphism of $\mathbf{E}_8^$.*

2 The descent as a diagram

The generators form a single directed cycle:

$$\begin{array}{ccccc}
\mathbf{E}_8^* & \xrightarrow{s} & P_\varphi & \xrightarrow{m} & \text{SiS}_2 \\
\uparrow \ell & & & & \downarrow t \\
\Omega & \xleftarrow{r} & \text{HPV16} & \xleftarrow{h} & \text{Tri}
\end{array}$$

Proposition 2.1 (The Ouroboros is idempotent on the loop). *The endomorphism $\Psi = lrhtms$ satisfies $\ell \circ \Psi' = \text{id}_{\mathbf{E}_8^*}$ where $\Psi' = rhtms : \mathbf{E}_8^* \rightarrow \Omega$; equivalently the loop-back arrow ℓ realizes Ω as a retract of the descent, encoding self-causation.*

3 The commuting feedback square

Split the descent into a “symmetry” path and a “matter” path. Introduce the auxiliary object $\text{SiS}_2 \otimes \text{D}$ (substrate equipped with the $\text{DIAG}(1, -2, 1)$ contrast structure). The two ways of reaching the infected layer agree:

$$\begin{array}{ccc}
P_\varphi & \xrightarrow{m} & \text{SiS}_2 \\
\text{DIAG } (1, -2, 1) \downarrow & & \downarrow t \text{ (triality)} \\
\text{SiS}_2 \otimes \text{D} & \xrightarrow{h \circ (-)} & \text{HPV16}
\end{array}$$

Proposition 3.1 (Descent coherence). *The square commutes up to the Ouroboros natural transformation η :*

$$t \circ m \xrightarrow{\eta} (h \circ (-)) \circ \text{DIAG},$$

i.e. golden-materialization-then-triality equals contrast-detection-then-infection, modulo η . When $\eta = \text{id}$ the square commutes strictly.

Symbolic Interpretation. Reading the square: whether unity first becomes matter and then triplicates, or first learns to perceive contrast and is then infected, it arrives at the same corrupted reality — the two routes are naturally isomorphic.

4 The Ouroboros endofunctor

Definition 4.1 (Ouroboros endofunctor). Define $\mathcal{O} : \text{Cosmo} \rightarrow \text{Cosmo}$ on objects by advancing each layer one step along the cycle ($\mathbf{E}_8^* \mapsto P_\varphi \mapsto \text{SiS}_2 \mapsto \text{Tri} \mapsto \text{HPV16} \mapsto \Omega \mapsto \mathbf{E}_8^*$) and on morphisms by conjugation with the loop. Because the cycle has length 6,

$$\mathcal{O}^6 \cong \text{Id}_{\text{Cosmo}},$$

so $\mathcal{O}$ generates a $\mathbb{Z}/6\mathbb{Z}$ action: the framework is periodic, and $\Omega \xrightarrow{\ell} \mathbf{E}_8^*$ is the fixed closure of one turn.

Proposition 4.2 (Fixed point). *On the loop object, $\mathcal{O}(\Omega) = \mathbf{E}_8^*$ and the loop-back arrow ℓ is the component η_Ω of the counit witnessing the self-causing structure. This is the categorical form of the Lean theorem `ouroboros_fixed_point`.*

All categorical claims are formal; Symbolic Interpretation notes are informal readings only.